AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

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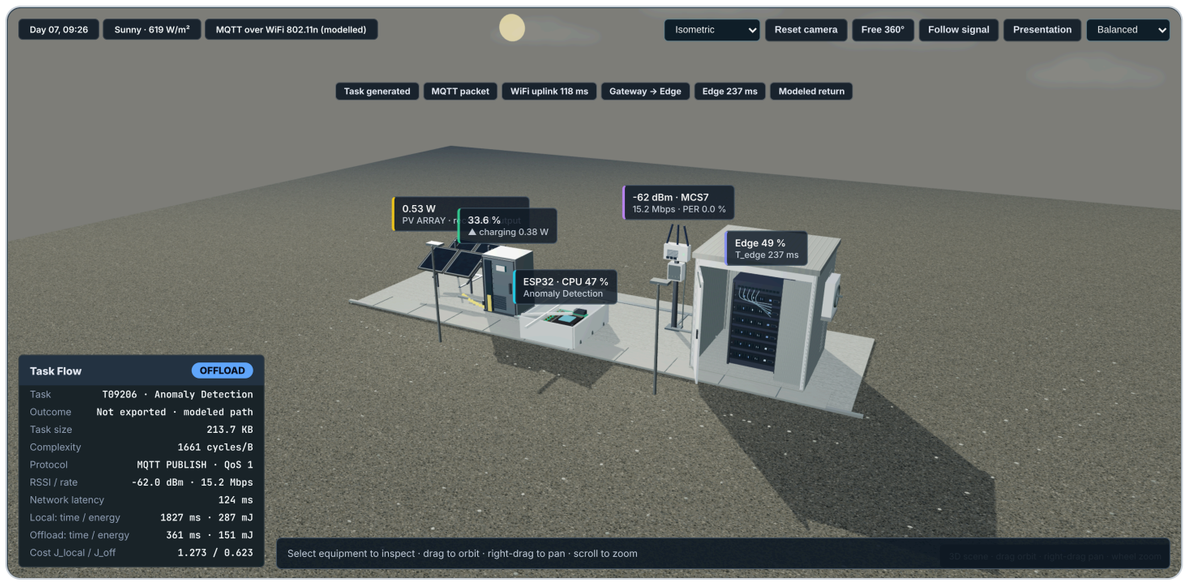
Department of Electrical and Electronics Engineering

19CSE362 · Fundamentals of Soft Computing

Methodology and Results

Four soft-computing algorithms, one explainable decision

Digital twin and dataset



The project's own 3D twin replaying task T09206 (day 7, 09:26, sunny): local 1.8 s and 287 mJ, offload 0.36 s and 151 mJ, so it is offloaded.

Solar panel
power = 1 W × sunlight ÷ 1000 × 0.85

Battery (12 Wh)
goes up with solar, down with power used

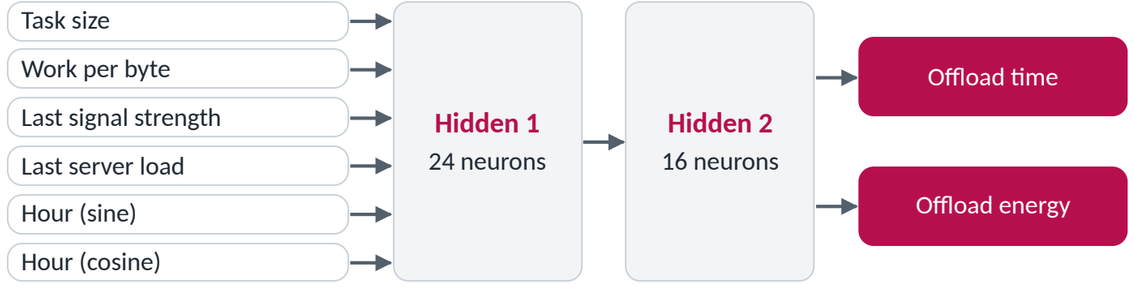
ESP32 (240 MHz)
work = data size × work per byte

WiFi and edge server
weaker signal → slower link; a busy server is slower

Dataset
10,080 tasks × 38 features, labelled with the cheaper option

The four soft-computing algorithms

Algorithm 1 — ANN multilayer perceptron



Accuracy R^2 **0.989** (time) and **0.987** (energy), median error about 5 %. Ablation: without task size R^2 falls to 0.924; without the last signal, to 0.968.

As a classifier on its own ("ANN only"): 96.7 %, but a black box

Algorithm 2 — Fuzzy logic Mamdani inference

Inputs battery, task load, WiFi, server → **Fuzzify** Low · Medium · High (trapezoids) → **12 rules** written by experts; AND = smallest → **Combine** take the largest of each output word → **Centroid** centre of the shape = offload score

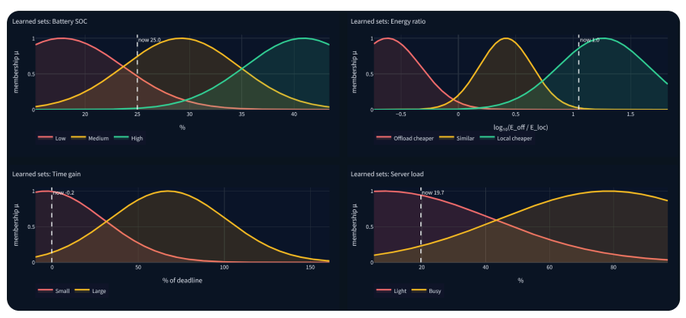
Example rule: **IF** task load is Heavy **AND** WiFi link is Good **THEN** OFFLOAD

Fully explainable and needs no training data, but the thresholds are an expert's guesses.

On its own ("Fuzzy only"): 92.5 % agreement

Algorithm 3 — ANFIS adaptive neuro-fuzzy system

1 · Fuzzify bell-shaped curves → **2 · Rules** multiply (AND) → **3 · Share** normalise → **4 · Output** straight-line formula → **5 · Add up** offload score



36 rules = 3 × 3 × 2 × 2
200 learned numbers
30 rounds of least squares + gradient descent

On its own, with raw inputs: 93.7 % agreement

Algorithm 4 — ANN + ANFIS hybrid proposed

Node knows battery, task, last signal, last load → **Algorithm 1** ANN predicts offload time and energy → **Algorithm 3** ANFIS 36 rules reason about it → **Decision** score above 0.5 → OFFLOAD

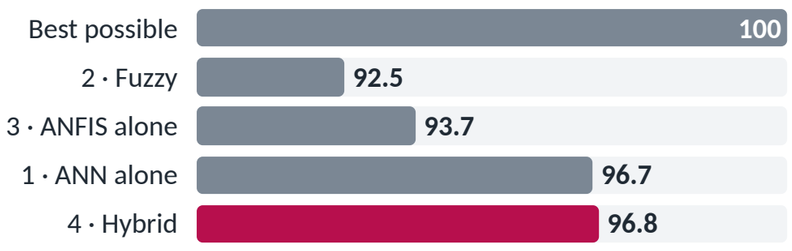
Task T09206: the ANN predicts 353 ms and 149 mJ (real 361 ms, 151 mJ). The strongest rule is *Battery Medium AND Offload cheaper AND Small gain AND Busy server*, so the score is 0.9965 → **OFFLOAD**.

ANFIS is trained on the ANN's predictions, so it learns to live with the ANN's errors.

96.8 % agreement · lowest energy · explains every decision

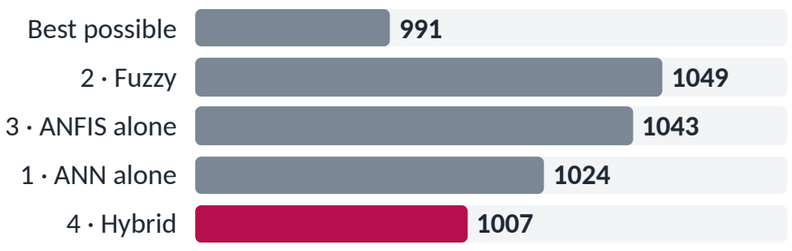
Results on 11 unseen weeks

Agreement with the best decision (%)



Bars run from 90 to 100 %. Higher is better.

Task energy per week (J)



Bars run from 950 to 1075 J. Lower is better.

Energy: the hybrid uses less than each algorithm on its own in all 11 weeks (Wilcoxon $p \approx 0.001$).

Accuracy: it beats Fuzzy and ANFIS alone in 11/11 weeks and ties the ANN ($p = 0.83$), while explaining itself.